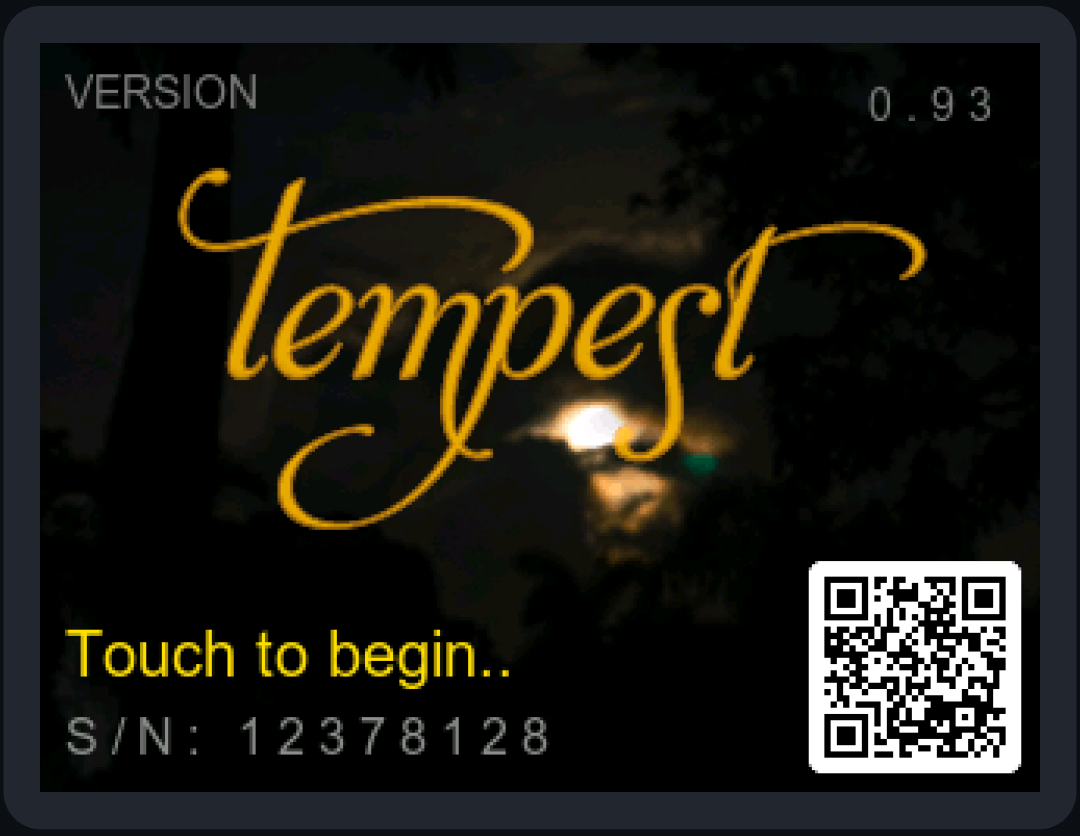
USER MANUAL

TEMPEST 2



Environmental monitor · NRG word engine · real-speech playback · E-field meter

Firmware v0.93

WELCOME

About this manual

TEMPEST 2 monitors its surroundings (temperature, humidity, dew point, air pressure and electric field) and runs the NRG word engine, which picks and speaks words at random moments influenced by changes in those readings. Every screen picture in this manual was captured directly from a TEMPEST 2 running firmware v0.93.

SECTION	WHAT'S IN IT
Getting started	Turning it on and off, charging, the start-up screen
Getting around	The three touch buttons and swiping between screens
Enviro screen	Live readings, clock, battery, and the Temp Change Alert
NRG Speech screen	The word engine, its seven modes, CLEAR and MODE
E Field screen	The electric-field meter
Settings	Date and time, volume, brightness, temperature calibration
SD card & data log	Card setup, the speech files, and the session log
Updating the firmware	Installing new firmware from your browser or the SD card
Troubleshooting	Fixes for common problems

GETTING STARTED

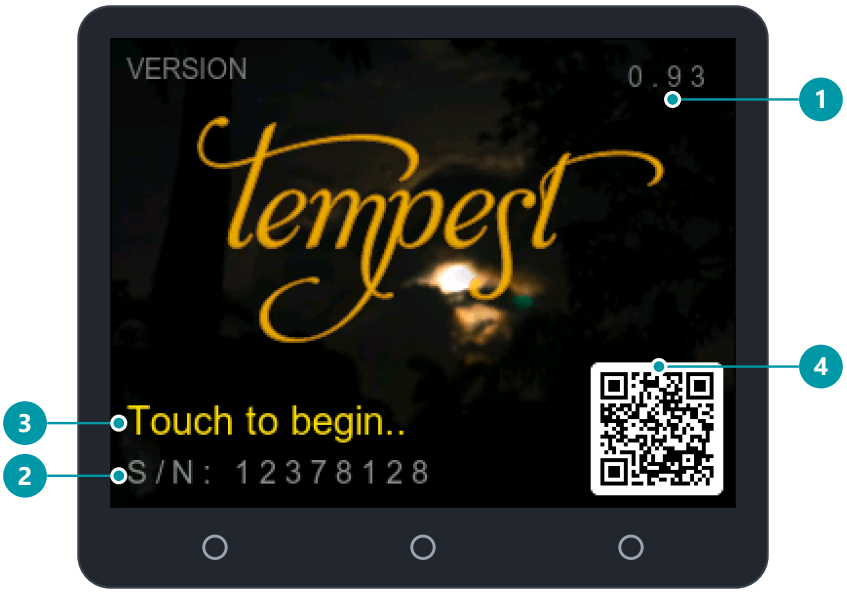
Turning TEMPEST 2 on

TEMPEST 2 runs on its built-in rechargeable battery and charges over USB-C.

Power

1. **To turn it on,** press the power button on the side of the unit. The screen lights up and the LED light bar starts a flickering flame effect.
2. **To turn it off,** hold the power button for about **6 seconds** until the screen goes dark.
3. **To charge,** plug in a USB-C cable. A lightning bolt appears on the battery indicator while it's on USB power.

The start-up screen



1. **Firmware version.** The version installed on this unit. Check it after an update.
2. **Serial number.** Unique to your unit. Quote it if you contact support.
3. **Touch to begin.** Tap anywhere on the screen to open the Enviro screen.
4. **QR code.** Scan it with your phone's camera to open the TEMPEST 2 web page, with this manual, firmware updates and the SD card files.

> If the SD card holds a firmware update file, the unit installs it at this point and shows a blue **Updating Tempest Firmware** screen. Don't turn it off until it finishes. See Updating the firmware.

GETTING AROUND

Moving between screens

TEMPEST 2 has three main screens. Use the touch buttons below the screen, or swipe.

Swipe left or right on any of these three screens to move to the next one (they wrap around).



ENVIRO
Left button

NRG SPEECH
Middle button

E FIELD
Right button

Tap a touch button below the screen to jump straight to that screen.

The touch buttons

Below the display are three touch areas, marked with small circles. They aren't physical buttons: just tap the area under the screen. The names along the bottom edge of the display show which screen each one opens. The current screen's name is shown in **green**.

TOUCH AREA	OPENS
Left	**ENVIRO**: environmental readings, clock and Temp Change Alert
Middle	**NRG SPEECH**: the word engine
Right	**E FIELD**: the electric-field meter

> Each touch area covers a third of the strip below the screen. If a tap doesn't register, tap a little higher, just under the edge of the display. Each tap gives a short buzz.

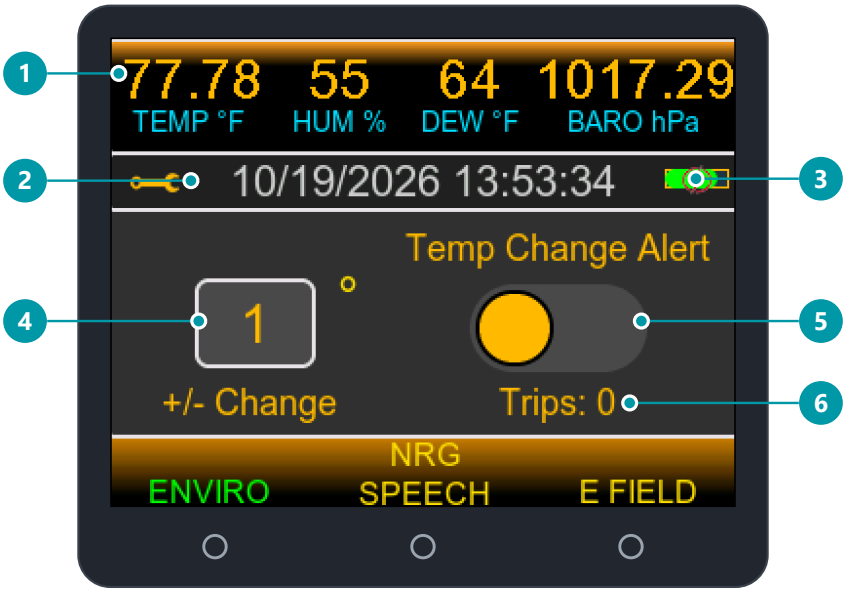
Readings on every main screen

The top band of the Enviro, NRG Speech and E Field screens always shows the live readings: **temperature** (°F), **humidity** (%), **dew point** (°F) and **barometric pressure** (hPa). They update about once a second. A battery indicator sits at the right of each screen.

SCREEN 1

Enviro

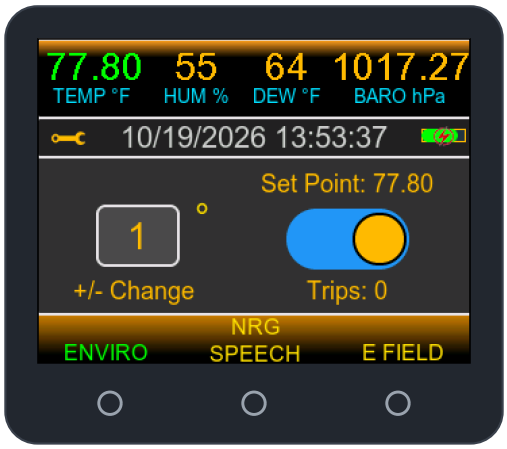
Live environmental readings, the clock, and an alert that watches for temperature changes.



1. **Live readings.** Temperature, humidity, dew point and barometric pressure.
2. **Date and time.** Tap the clock or the wrench to open **Settings**.
3. **Battery.** Charge level. The lightning bolt shows while it's on USB power.
4. **+/- Change.** How many degrees the temperature must move to trip the alert. Scroll to choose.
5. **Temp Change Alert switch.** Turns the alert on and off.
6. **Trips.** How many times the alert has tripped since you turned it on.

Using the Temp Change Alert

1. Scroll the **+/- Change** box to the number of degrees you want to watch for.
2. Tap the **Temp Change Alert** switch to turn it on. TEMPEST takes the current temperature as the **Set Point**, shows it above the switch, and locks the +/- Change box.
3. The temperature turns **green** while it stays within your range of the set point.
4. If it moves that many degrees or more above or below the set point, the temperature turns **red** and **Trips** goes up by one. It turns green again once the temperature comes back.
5. Tap the switch again to turn the alert off. That clears the set point and resets Trips to 0.

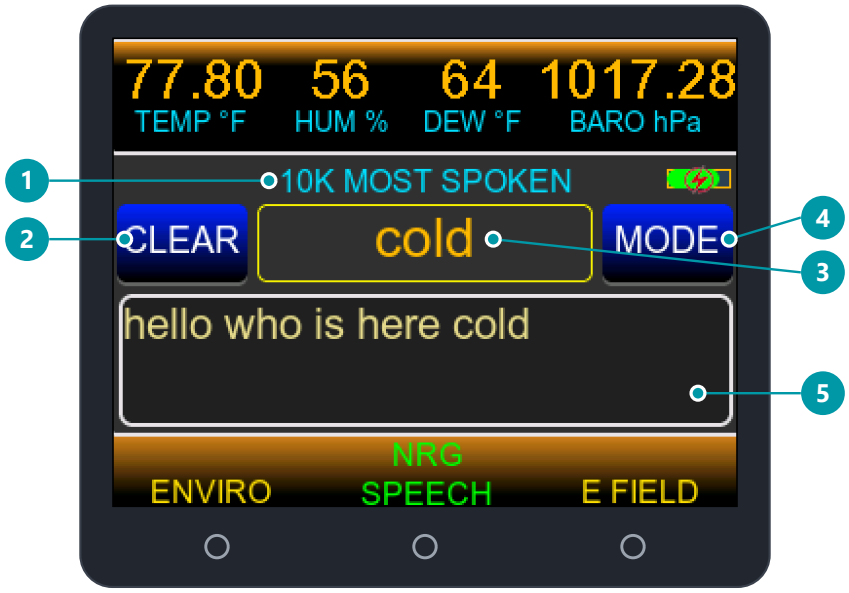


Alert on, within range

SCREEN 2

NRG Speech

The NRG word engine picks words at random moments and shows and speaks them.



1. **Current mode.** Which word list is in use. See the table below.
2. **CLEAR.** Clears the current word and the history.
3. **Current word.** The most recent word, in large type.
4. **MODE.** Steps to the next mode each time you tap it.
5. **History.** Recent words, oldest first, so you can read them as a sentence.

How words arrive

Words only arrive while the **NRG Speech** screen is showing. They come at random moments. In calm conditions that's typically every 20–30 seconds, sometimes longer, and now and then two or three arrive close together. Sudden changes make words come **more often**: a jump in the electric-field reading (see E Field) or a quick swing in temperature.

When a word arrives, the unit gives a short buzz, shows the word, adds it to the history and speaks it.

> **Speech.** Modes 1 to 6 below are spoken by the optional **speech module**. Without it the words are still shown, just not spoken. **4.4K Real Speech** plays recorded voices through TEMPEST's own speaker and needs the speech files on the SD card. Set the loudness with **Volume** in Settings. In 4.4K Real Speech mode, tapping a touch button while a word plays stops it and changes screen straight away.

The seven modes

MODE (AS SHOWN)	WORDS USED	NEEDS
10K MOST SPOKEN (10K ENGLISH)	The 10,000 most commonly used English words	Speech module to hear it
29K WRD & PHRASES	A large list of English words and short phrases	Speech module to hear it
PHRASES MODE	133 complete phrases	Speech module to hear it
ALPHA MODE	Single letters, A to Z	Speech module to hear it
10K SPANISH	The 10,000 most commonly used Spanish words	Speech module to hear it
CUSTOM SD CARD	Your own list from **customwords.txt** on the SD card	SD card; speech module to hear it
4.4K REAL SPEECH	4,409 recorded spoken words	SD card speech files (built-in speaker)

The first tap of **MODE** moves from 10K Most Spoken to 29K Words & Phrases. After 4.4K Real Speech it wraps back to 10K English.

Your own word list

For **CUSTOM SD CARD** mode, put a plain-text file named **customwords.txt** at the top level of the SD card, with one word or phrase per line: up to 2,000 lines, each up to 19 characters. The list loads the first time you select the mode, so restart TEMPEST after changing the file. If the screen shows **SD LIBRARY NOT FOUND**, the file is missing or misnamed.

SCREEN 3

E Field

A live meter for the electric field around the unit.



1. **Level bar.** Fills from left to right as the field gets stronger.
2. **Reading.** The current level in mV/m. It updates five times a second and is lightly smoothed.
3. **EFIELD LEVEL.** Label for the reading.

Using the meter

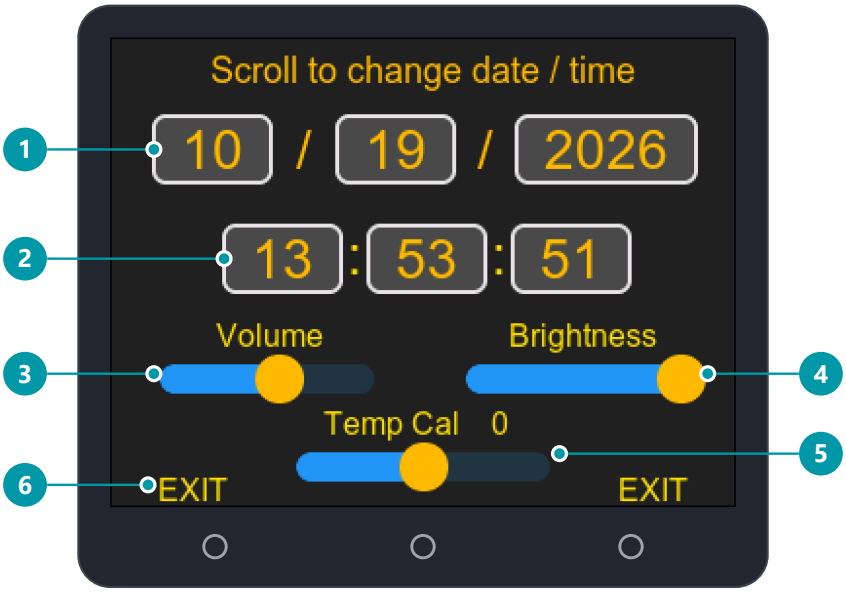
Hold the unit steady and watch for changes rather than any single number. Mains wiring, chargers, phones and other electronics all raise the reading. On the NRG Speech screen, sudden jumps make words arrive more often.

> The reading shown in this picture is a sample value. On a quiet bench, away from electronics, it usually sits near 0.

SETTINGS

Date, time and preferences

Open Settings by tapping the **date and time** or the **wrench** on the Enviro screen.



1. **Date.** Month / day / year.
2. **Time.** Hours : minutes : seconds, in 24-hour time.
3. **Volume.** Speech loudness, 0 (quietest) to 9.
4. **Brightness.** Screen brightness. Slide left to dim it and save battery.
5. **Temp Cal.** Shifts the temperature reading by -4 to +4 °F to match a reference thermometer.
6. **EXIT.** Tap the left or right touch button below the screen to leave Settings.

Setting the clock

1. Scroll each box until it shows the value you want.
2. Let go: the clock is set to the date and time shown as soon as you release a box.
3. Tap the left or right touch button to leave. The new time appears on the Enviro screen.

> The clock keeps time while the unit is off. **Volume**, **Brightness** and **Temp Cal** go back to their defaults (5, full brightness and 0) each time TEMPEST is turned on.

SD CARD & DATA LOG

The microSD card

The SD card holds your custom word list and the 4.4K Real Speech recordings, and TEMPEST records each session on it.

Choosing and preparing a card

1. Use a **name-brand** microSD card (SanDisk, Samsung or Kingston). **32 GB or smaller** is easiest.
2. Format it as **FAT32**. exFAT, the usual format for 64 GB and larger cards, won't work.
3. Download the TEMPEST 2 SD card files from **ozinfl.github.io/tempest2-flasher**, open the zip, and copy everything inside it to the top level of the card.
4. With TEMPEST off, push the card fully into the slot until it clicks, then turn it on.

> **Generic, no-brand cards are known to fail in TEMPEST 2.** They work for a moment after power-on, then stop responding once the screen is in use. That shows up as **NO SD CARD MOUNTED** or a silent 4.4K Real Speech mode. Reflashing won't fix it; a different card will.

What goes on the card

FILE OR FOLDER	WHAT IT'S FOR
customwords.txt	Your word list for CUSTOM SD CARD mode
af/ (1.wav … 4409.wav)	The recordings for 4.4K REAL SPEECH mode
TEMPESTdata.txt	Created by TEMPEST: your session log (see below)
firmware.bin	Only when updating firmware from the card. TEMPEST deletes it after installing.

The session log

Each time TEMPEST starts, it writes a **New Session Started** header to **TEMPESTdata.txt**, then adds one line per second. Each line holds the date and time, the screen you were on, temperature, humidity, dew point, pressure, the E-field level, the latest NRG word, and the last six words as a sentence. The values are comma-separated, so the file opens in Excel or any spreadsheet app.

Lines are saved to the card in batches every 10 seconds, so turning the unit off can lose the last few seconds of a session. If the card stops accepting data, TEMPEST stops logging for that session rather than slowing down; the readings and words carry on as normal.

Installing new firmware

Check your current version on the start-up screen.

From your web browser (recommended)

1. On a Windows, Mac or Linux computer, open **ozinfl.github.io/tempest2-flasher** in **Chrome** or **Edge**.
2. Turn TEMPEST on and connect it with a USB-C **data** cable (some cables only charge).
3. Click **Flash TEMPEST 2**, choose the port (usually named **CP210x**, **USB Serial** or **CH9102**) and click **Install**.
4. Wait about a minute until it says it's done. TEMPEST restarts and the start-up screen shows the new version.

> **No port in the list?** Close any other program that uses serial ports, try another cable or USB port, and on Windows install the CP210x or CH9102 USB driver (both are linked on the flasher page).

From the SD card

1. Download **firmware.bin** from the flasher page and copy it to the top level of the SD card. Keep the name **firmware.bin**.
2. Put the card in TEMPEST and turn it on. A blue **Updating Tempest Firmware** screen appears. **Don't turn it off.**
3. When it finishes, TEMPEST deletes firmware.bin from the card. Restart the unit to run the new version.

TROUBLESHOOTING

Common problems

PROBLEM	WHAT TO DO
Screen stays black after an update	Flash it again from the flasher page; that fixes it. If the unit won't respond at all, hold the power button 6 seconds, then press it once to turn it back on.
NO SD CARD MOUNTED	Reseat the card with the unit off. If it keeps happening, use a name-brand FAT32 card (see SD card).
SD LIBRARY NOT FOUND	The card works, but **customwords.txt** isn't at its top level.
4.4K Real Speech is silent	Check that the **af** folder is on the card and Volume isn't at 0. If the card isn't working, see above.
Words show but aren't spoken	Every mode except 4.4K Real Speech is spoken by the optional speech module. Check it's connected, or use 4.4K Real Speech.
A touch button doesn't respond	Tap just below the edge of the display, in the middle of that third.
Wrong date or time	Set it in **Settings** (tap the clock on the Enviro screen).
Temperature reads high or low	Adjust **Temp Cal** in Settings to match a reference thermometer.
The flasher says Failed to open serial port	Another program has the port open. Close serial monitors and similar apps, then try again.